Closed Discrete Selection and the Compact-open Topology

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Basic Notions and Notation

For sets $\mathcal{A}$ and $\mathcal{B}$, we define **selection principles**.

$S_1(\mathcal{A}, \mathcal{B})$

For every $\{A_n : n \in \omega\} \subseteq \mathcal{A}$, there exists $\{B_n : n \in \omega\}$ so that

$$(\forall n \in \omega)[B_n \in A_n] \text{ and } \{B_n : n \in \omega\} \in \mathcal{B}.$$

$S_{\text{fin}}(\mathcal{A}, \mathcal{B})$

For every $\{A_n : n \in \omega\} \subseteq \mathcal{A}$, there exists $\{B_n : n \in \omega\}$ so that

$$(\forall n \in \omega)[B_n \in [A_n]^{<\omega}] \text{ and } \bigcup\{B_n : n \in \omega\} \in \mathcal{B}.$$

For a topological space X,

- $\mathscr{T}_X$ — all non-empty open sets
- CD_X — all closed and discrete subsets

Closed Discrete Selection (Tkachuk)

$$S_1(\mathscr{T}_X, \mathrm{CD}_X)$$

Theorem (Tkachuk, 2017)

For a Tychonoff (Hausdorff and completely regular) space X, the following are equivalent:

- $C_p(X) \models \neg S_1(\mathscr{T}_{C_p(X)}, \mathrm{CD}_{C_p(X)})$
- $C_p(X)$ is first-countable (equivalently, metrizable)
- X is countable

Selection principles naturally extend to **selection games**.

$G_1(\mathcal{A}, \mathcal{B})$

In the nth inning, One chooses $A_n \in \mathcal{A}$ and Two chooses $B_n \in A_n$.

- Two is declared the winner if $\{B_n : n \in \omega\} \in \mathcal{B}$.

$G_{\text{fin}}(\mathcal{A}, \mathcal{B})$

In the nth inning, One chooses $A_n \in \mathcal{A}$ and Two chooses $B_n \in [A_n]^{<\omega}$.

- Two is declared the winner if $\bigcup\{B_n : n \in \omega\} \in \mathcal{B}$.

Let $\mathcal{G}$ be a game.

Perfect Information Strategies

- A **winning strategy** for One (resp. Two) is any rule which dictates One's (resp. Two's) next move based on all of the previous plays and results in a win for One (resp. Two).
  - $\mathrm{I} \uparrow \mathcal{G}$
  - $\mathrm{II} \uparrow \mathcal{G}$

Limited Information Strategies

- A **pre-determined** winning strategy for One only depends on the current turn number.
  - $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathcal{G}$
- A **tactic** for One only depends on Two's most recent play.
  - $\mathrm{I} \underset{\text{tact}}{\uparrow} \mathcal{G}$
- A **Markov** strategy for Two only depends on the current turn number and One's most recent play.
  - $\mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G}$

For two games $\mathcal{G}_1$ and $\mathcal{G}_2$, we have notions of duality.

Strategically Dual

$$(\mathrm{I} \uparrow \mathcal{G}_1 \Leftrightarrow \mathrm{II} \uparrow \mathcal{G}_2) \text{ and } (\mathrm{I} \uparrow \mathcal{G}_2 \Leftrightarrow \mathrm{II} \uparrow \mathcal{G}_1)$$

Markov Dual

$$(\mathrm{I} \underset{\text{pre}}{\uparrow} \mathcal{G}_1 \Leftrightarrow \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G}_2) \text{ and } (\mathrm{I} \underset{\text{pre}}{\uparrow} \mathcal{G}_2 \Leftrightarrow \mathrm{II} \underset{\text{mark}}{\uparrow} \mathcal{G}_1)$$

Dual

Both strategically and Markov dual.

Fact

$X \models S_1(\mathcal{A}, \mathcal{B})$ if and only if $\mathrm{I} \underset{\text{pre}}{\not\uparrow} G_1(\mathcal{A}, \mathcal{B})$.

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So we can study selection principles using games and limited information strategies.

For a topological space X,

- $\mathcal{O}_X$ — the open covers of X
- $\mathscr{N}(A)$ — all open sets containing A
- $\mathscr{N}[\mathcal{A}]$ — $\{\mathscr{N}(A) : A \in \mathcal{A}\}$

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Theorem (Tkachuk, 2017)

For a Tychonoff space X, the following are equivalent:

- $\mathrm{I} \underset{\text{pre}}{\uparrow} G_1(\mathscr{T}_{C_p(X)}, \mathrm{CD}_{C_p(X)})$
- $C_p(X)$ is first-countable (equivalently, metrizable)
- X is countable
- $\mathrm{I} \underset{\text{pre}}{\uparrow} G_1(\mathscr{N}[X], \neg\mathcal{O}_X)$

We move to $C_k(X)$ and compact sets.

- $K(X)$ — the non-empty compact subsets

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- $K(X)$ — the non-empty compact subsets

Compact-open Game

In the nth inning, One chooses a compact set K_n and Two chooses an open set $U_n \supseteq K_n$.

- One is declared the winner if Two covers the space.

Compact-open Game

$$G_1(\mathscr{N}[K(X)], \neg\mathcal{O}_X)$$

Compact-open Game

$$G_1(\mathscr{N}[K(X)], \neg\mathcal{O}_X)$$

Fact (à la Telgárski)

If every compact subset of X is a G_δ, then X is σ-compact if and only if $\mathrm{I} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{O}_X)$.

Compact-open Game

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Fact (à la Telgárski)

A Hausdorff space X is σ-compact if and only if $\mathrm{I} \underset{\text{pre}}{\uparrow} G_1(\mathscr{N}[K(X)], \neg\mathcal{O}_X)$.

Is there a relationship between $G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$ and $G_1(\mathscr{N}[K(X)], \neg\mathcal{O}_X)$?

Definition

A space X is said to be **hemicompact** if there exists a sequence $\{K_n : n \in \omega\}$ of compact sets so that every compact set is contained in some K_n.

Definition

For a space X, an (infinite) open cover $\mathscr{U}$ of X is said to be

- a **k-cover** if every compact set is contained in a member of $\mathscr{U}$
  - $\mathcal{K}_X$ — the k-covers of X
- a **γ_k-cover** if every compact set is contained in all but finitely many members of $\mathscr{U}$
  - $\Gamma_k(X)$ — the γ_k-covers of X

Results

Theorem (C., Holshouser)

For a Tychonoff space X, the following are equivalent:

- $\mathrm{I} \underset{\mathrm{pre}}{\uparrow} G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$
- $C_k(X)$ is first-countable (equivalently, metrizable)
- X is hemicompact
- $\mathbb{K}(X)$ is hemicompact
- $\mathbb{K}(X)$ is σ-compact
- $\mathrm{I} \underset{\mathrm{pre}}{\uparrow} G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$

- Suppose $\mathrm{I} \underset{\text{pre}}{\uparrow} G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$.

- Suppose I $\underset{\text{pre}}{\uparrow}$ $G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$.
- Let $\{\langle f_n, K_n, \varepsilon_n \rangle : n \in \omega\}$ witness the pre-determined winning strategy for One.

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- Let $\{\langle f_n, K_n, \varepsilon_n\rangle : n \in \omega\}$ witness the pre-determined winning strategy for One.
- Suppose K is compact so that $K \not\subseteq K_n$ for any $n \in \omega$.
- For $n \in \omega$, let $x_n \in K \setminus K_n$ and let $g_n \in C_k(X)$ satisfy $g_n \restriction_{K_n} \equiv f_n \restriction_{K_n}$ and $g_n(x_n) = n$.

- Suppose I $\underset{\text{pre}}{\uparrow}$ $G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$.
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- The $\{g_n : n \in \omega\}$ contradicts the winning strategy for One.

What about perfect information strategies?

What about perfect information strategies?

For a space X and a set $\mathcal{A}$,

- $\mathcal{O}(X, \mathcal{A})$ — all open covers $\mathscr{U}$ so that, for all $A \in \mathcal{A}$, there exists $U \in \mathscr{U}$ so that $A \subseteq U$
- $\Gamma(X, \mathcal{A})$ — all open covers $\mathscr{U}$ so that, for all $A \in \mathcal{A}$, $A \subseteq U$ for all but finitely many $U \in \mathscr{U}$

Definition (Clontz)

- For a set $\mathcal{A}$, let

$$\mathbf{C}(\mathcal{A}) = \left\{ f : \mathcal{A} \to \bigcup \mathcal{A} | f(A) \in A \right\}.$$

- A set $\mathcal{B} \subseteq \mathcal{A}$ is a **selection basis** if

$$(\forall A \in \mathcal{A})(\exists B \in \mathcal{B})[B \subseteq A].$$

- Then a set $\mathcal{R}$ is a **reflection** of $\mathcal{A}$ if

$$\{f[\mathcal{R}] : f \in \mathbf{C}(\mathcal{R})\}$$

is a selection basis for $\mathcal{A}$.

Theorem (Clontz, 2018)

If $\mathcal{R}$ is a reflection of $\mathcal{A}$, then $G_1(\mathcal{A}, \mathcal{B})$ and $G_1(\mathcal{R}, \neg\mathcal{B})$ are dual for any set $\mathcal{B}$.

Theorem (Clontz, 2018)

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Proposition

For a space X and a set $\mathcal{A}$ of subsets of X, $\mathscr{N}[\mathcal{A}]$ is a reflection of $\mathcal{O}(X, \mathcal{A})$.

Corollary

For a space X and a set $\mathcal{A}$ of subsets of X, $G_1(\mathcal{O}(X, \mathcal{A}), \mathcal{B})$ and $G_1(\mathscr{N}[\mathcal{A}], \neg\mathcal{B})$ are dual for any set $\mathcal{B}$.

Theorem (C., Holshouser)

For a Tychonoff space X, the following are equivalent:

- $\mathrm{I} \uparrow G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$
- $\mathrm{I} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$
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For a Tychonoff space X, the following are equivalent:

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- $\mathrm{I} \underset{\text{pre}}{\uparrow} G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$
- $\mathrm{II} \underset{\text{mark}}{\uparrow} G_1(\mathcal{K}_X, \mathcal{K}_X)$

Example

The one-point Lindelöfication X of ω_1 with the discrete topology is a space so that

$$\mathrm{I} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$$

yet

$$\mathrm{I} \underset{\text{pre}}{\not\uparrow} G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$$

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X is not hemicompact.

$\mathrm{I} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$

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- Two plays a co-countable set containing ω_1; equivalently, a set $A_0 = \{x_{0,n} : n \in \omega\}$ where $A_0 \cap K_0 = \emptyset$.

$\mathrm{I} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$

- One opens with $K_0 = \{\omega_1\}$.
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- One plays $K_1 = \{\omega_1, x_{0,0}\}$.

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- One plays $K_1 = \{\omega_1, x_{0,0}\}$.
- Two responds with $A_1 = \{x_{1,n} : n \in \omega\}$ with $A_1 \cap K_1 = \emptyset$.

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- One plays $K_1 = \{\omega_1, x_{0,0}\}$.
- Two responds with $A_1 = \{x_{1,n} : n \in \omega\}$ with $A_1 \cap K_1 = \emptyset$.
- One plays $K_2 = \{\omega_1, x_{0,0}, x_{0,1}, x_{1,0}, x_{1,1}\}$.

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- One plays $K_1 = \{\omega_1, x_{0,0}\}$.
- Two responds with $A_1 = \{x_{1,n} : n \in \omega\}$ with $A_1 \cap K_1 = \emptyset$.
- One plays $K_2 = \{\omega_1, x_{0,0}, x_{0,1}, x_{1,0}, x_{1,1}\}$.
- $\vdots$
- Two responds with $A_\ell = \{x_{\ell,n} : n \in \omega\}$ with $A_\ell \cap K_\ell = \emptyset$.

$\mathrm{I} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$

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- Two responds with $A_1 = \{x_{1,n} : n \in \omega\}$ with $A_1 \cap K_1 = \emptyset$.
- One plays $K_2 = \{\omega_1, x_{0,0}, x_{0,1}, x_{1,0}, x_{1,1}\}$.
- $\vdots$
- Two responds with $A_\ell = \{x_{\ell,n} : n \in \omega\}$ with $A_\ell \cap K_\ell = \emptyset$.
- One plays $K_{\ell+1} = \{\omega_1\} \cup \{x_{j,k} : j, k < \ell + 1\}$.

Theorem (Pawlikowski, 1994)

Both

$$\mathrm{I} \uparrow G_{\mathrm{fin}}(\mathcal{O}, \mathcal{O}) \Leftrightarrow \mathrm{I} \underset{\mathrm{pre}}{\uparrow} G_{\mathrm{fin}}(\mathcal{O}, \mathcal{O})$$

and

$$\mathrm{I} \uparrow G_1(\mathcal{O}, \mathcal{O}) \Leftrightarrow \mathrm{I} \underset{\mathrm{pre}}{\uparrow} G_1(\mathcal{O}, \mathcal{O}).$$

Theorem (C., Holshouser)

For $\mathcal{A} \subseteq \mathcal{B}$,

$$\mathrm{I} \uparrow G_{\mathrm{fin}}(\mathcal{O}(X,\mathcal{A}),\mathcal{O}(X,\mathcal{B})) \Leftrightarrow \mathrm{I} \underset{\mathrm{pre}}{\uparrow} G_{\mathrm{fin}}(\mathcal{O}(X,\mathcal{A}),\mathcal{O}(X,\mathcal{B}))$$

and

$$\mathrm{I} \uparrow G_{1}(\mathcal{O}(X,\mathcal{A}),\mathcal{O}(X,\mathcal{B})) \Leftrightarrow \mathrm{I} \underset{\mathrm{pre}}{\uparrow} G_{1}(\mathcal{O}(X,\mathcal{A}),\mathcal{O}(X,\mathcal{B})).$$

Theorem (C., Holshouser)

For a Tychonoff space X, the following are equivalent:

- $\mathrm{II} \uparrow G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$
- $\mathrm{II} \underset{\text{mark}}{\uparrow} G_1(\mathscr{T}_{C_k(X)}, \mathrm{CD}_{C_k(X)})$
- $\mathrm{I} \uparrow G_1(\mathcal{K}_X, \mathcal{K}_X)$
- $\mathrm{I} \underset{\text{pre}}{\uparrow} G_1(\mathcal{K}_X, \mathcal{K}_X)$
- $\mathrm{II} \uparrow G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$
- $\mathrm{II} \underset{\text{mark}}{\uparrow} G_1(\mathscr{N}[K(X)], \neg\mathcal{K}_X)$

Future Work

- How much of this can be recovered for $C_k(X, [0, 1])$?
- What about longer length games?

Thank You!